Deployment

NetApp Solutions

NetApp
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Deployment

Deploy Advanced Cluster Management for Kubernetes: Red Hat OpenShift with NetApp

Prerequisites

1. A Red Hat OpenShift cluster (greater than version 4.5) for hub cluster
2. Red Hat OpenShift clusters (greater than version 4.4.3) for managed clusters
3. Cluster-admin access to Red Hat OpenShift cluster
4. A Red Hat subscription for Advanced Cluster Management for Kubernetes

Advanced Cluster Management is an add-on on for the OpenShift cluster, so there are certain requirements and restrictions on the hardware resources based on the features used across the hub and managed clusters. You need to take these issues into account when sizing the clusters. See the documentation here for more details.

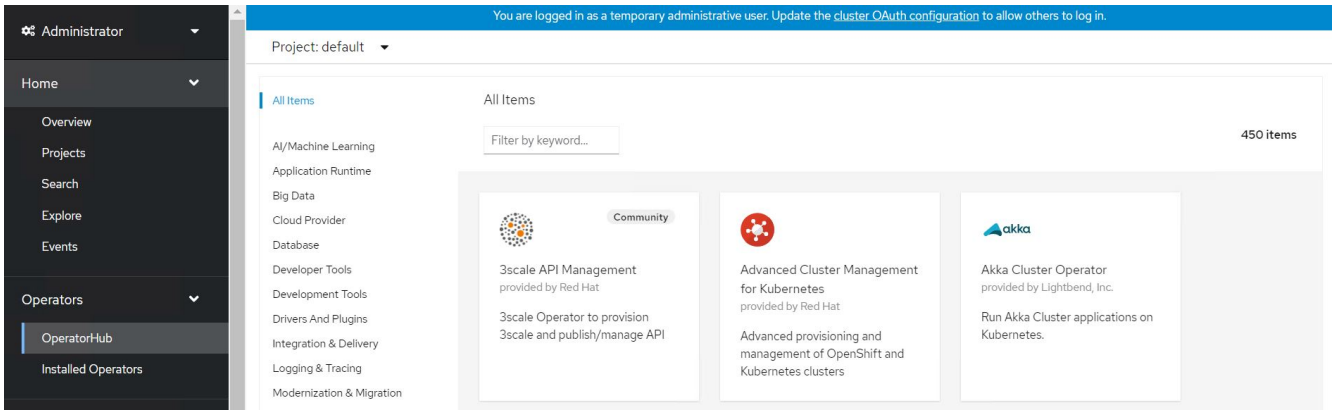
Optionally, if the hub cluster has dedicated nodes for hosting infrastructure components and you would like to install Advanced Cluster Management resources only on those nodes, you need to add tolerations and selectors to those nodes accordingly. For more details, see the documentation here.

Next: Installation.

Deploy Advanced Cluster Management for Kubernetes: Red Hat OpenShift with NetApp

To install Advanced Cluster Management for Kubernetes on an OpenShift cluster, complete the following steps:

1. Choose an OpenShift cluster as the hub cluster and log into it with cluster-admin privileges.
2. Navigate to `Operators` → `Operators Hub` and search for `Advanced Cluster Management for Kubernetes`.



3. Select the `Advanced Cluster Management for Kubernetes` and click `Install`.

4. On the Install Operator screen, provide the necessary details (NetApp recommends retaining the default parameters) and click `Install`.

OperatorHub > Operator Installation

Install Operator

Install your Operator by subscribing to one of the update channels to keep the Operator up to date. The strategy determines either manual or automatic updates.

Update channel *

- release-2.0
- release-2.1
- release-2.2

Installation mode *

- All namespaces on the cluster (default)
  This mode is not supported by this Operator
- A specific namespace on the cluster
  Operator will be available in a single Namespace only.

Installed Namespace *

- Operator recommended Namespace: PR open-cluster-management

Namespace creation
Namespace **open-cluster-management** does not exist and will be created.

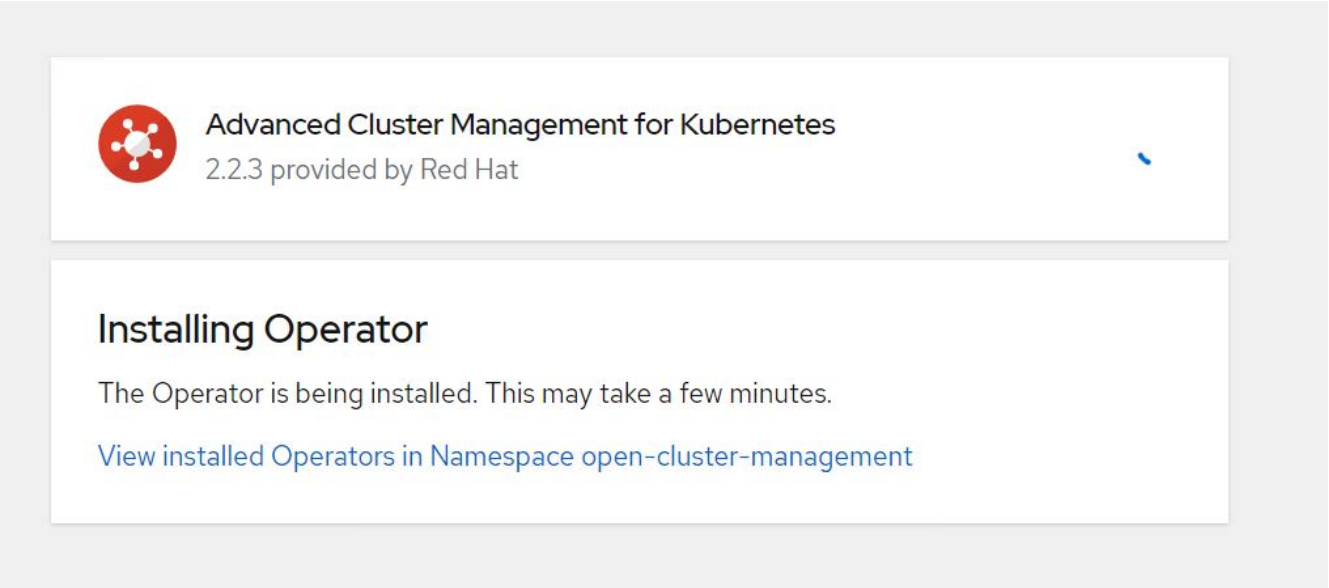
- Select a Namespace

Approval strategy *

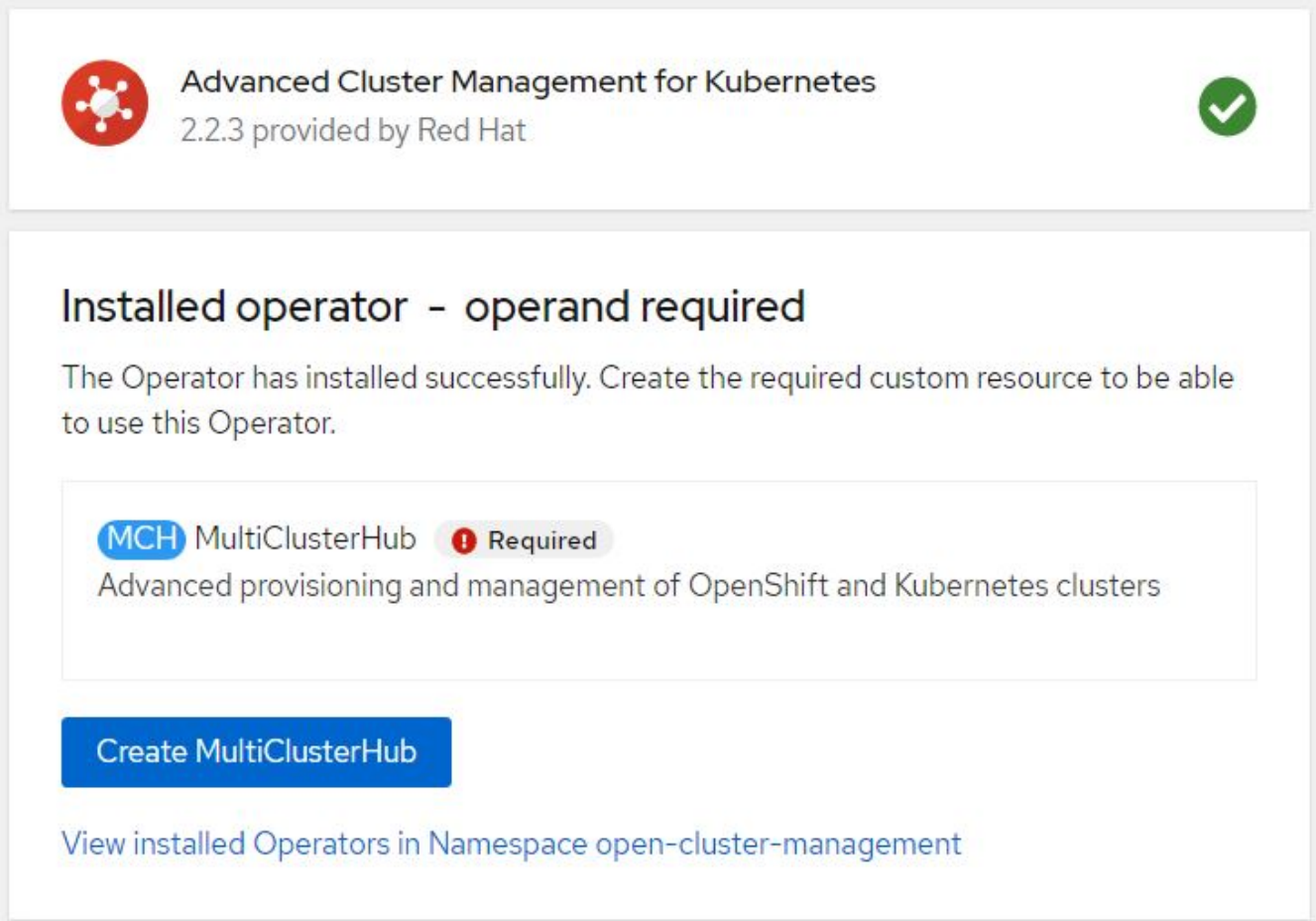
- Automatic
- Manual

Install Cancel

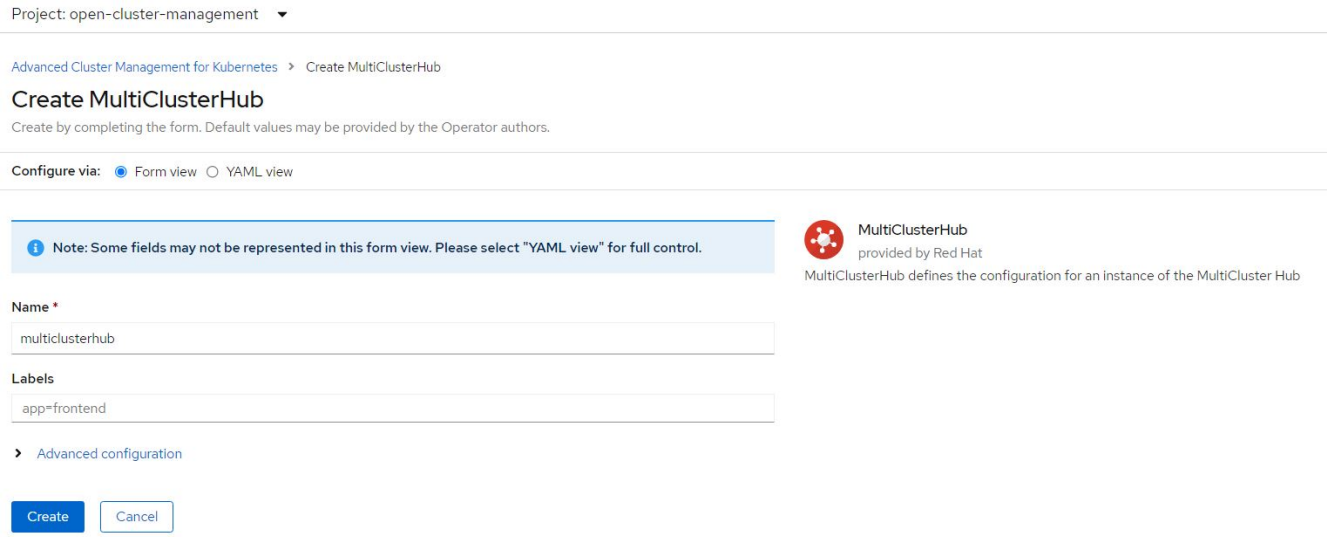
5. Wait for the operator installation to complete.



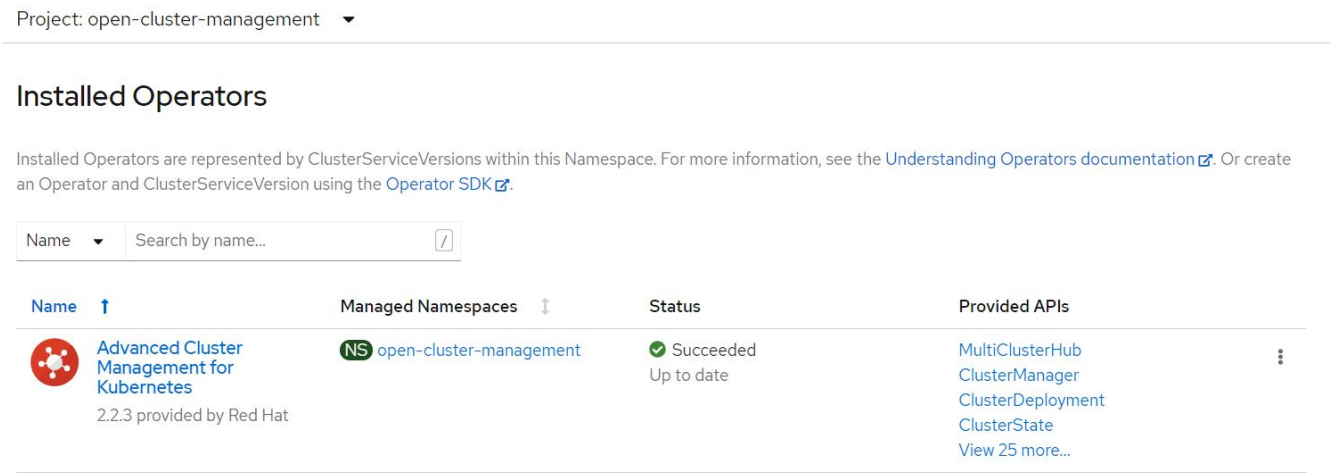
6. After the operator is installed, click `Create MultiClusterHub`.



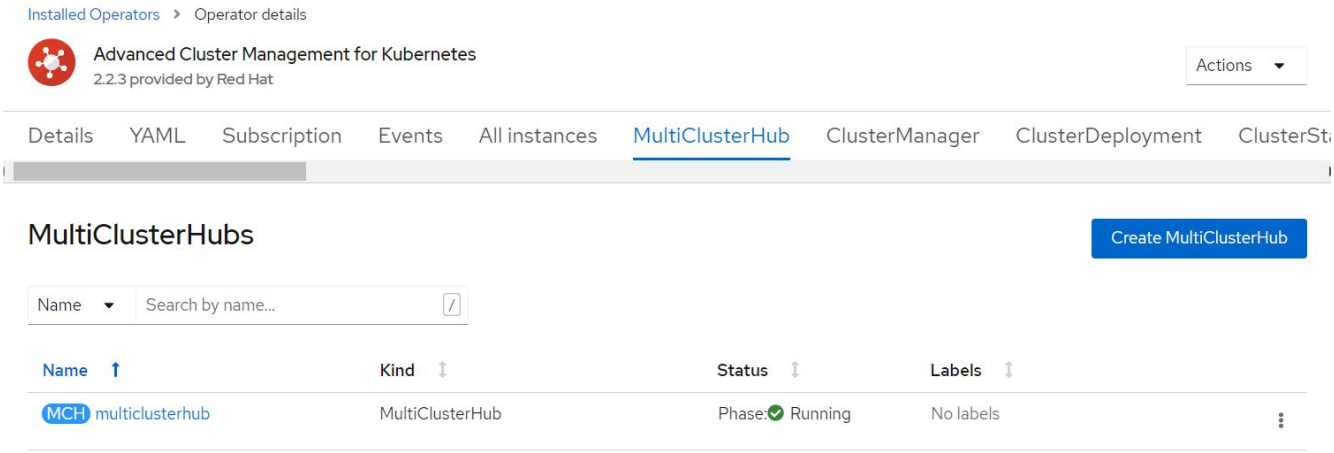
7. On the `Create MultiClusterHub` screen, click `Create` after furnishing the details. This initiates the installation of a multi-cluster hub.



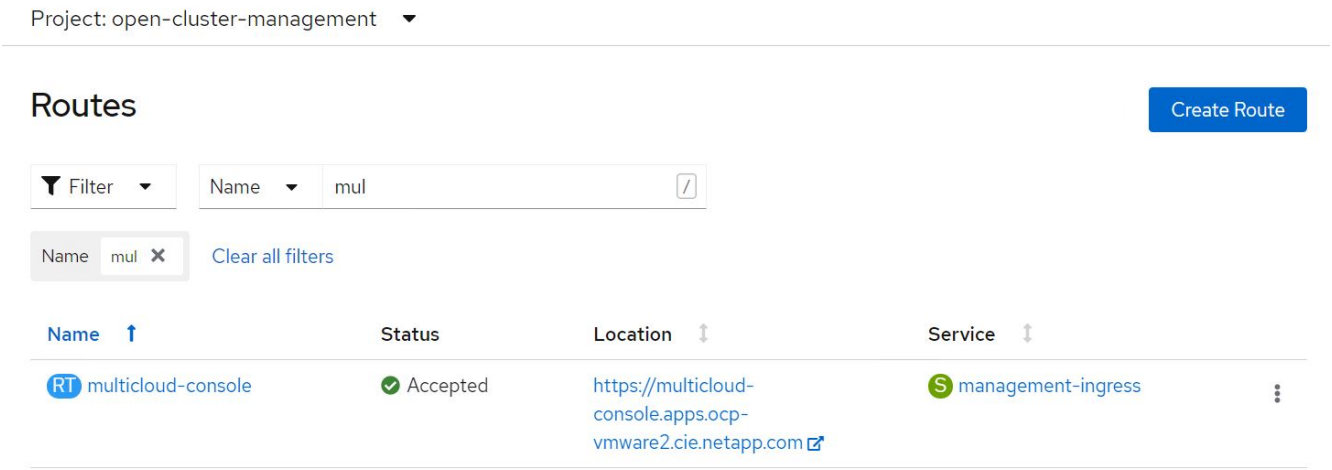
8. After all the pods move to the `Running` state in the open-cluster-management namespace and the operator moves to the `Succeeded` state, Advanced Cluster Management for Kubernetes is installed.



9. It takes some time to complete the hub installation, and, after it is done, the MultiCluster hub moves to `Running` state.



10. It creates a route in the open-cluster-management namespace, connect to the URL in the route to access the Advanced Cluster Management console.



Next: Features - Cluster Lifecycle Management.

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